

Is Banking Consolidation at Odds with Financial Stability?

The Case of India

TASNEEM CHHERAWALA, GARGI SANATI

This paper examines the nexus between consolidation and financial stability in the context of a series of restructuring mergers that have occurred in the Indian banking sector in recent years. Employing a system-GMM two-stage panel regression model and a bias-corrected random effect model for the period from March 2012 to June 2022, it concludes that mergers increase financial fragility, as measured by banks' marginal risk contribution to systemic risk. The destabilising effect is also driven by bank-specific risk factors, standalone insolvency risk, and ownership characteristics but is neutral to bank size. Furthermore, the build-up of good-quality capital for potential loss mitigation can alleviate the adverse impact on systemic risk. The results have important implications for macro-prudential policies.

The global financial crisis (GFC) of 2007–08 underscored the fact that micro-prudential regulations, which focused on the soundness of standalone financial institutions, were not sufficient to ensure the stability of the financial system as a whole. Losses incurred by stressed banks, which were systemically important, were transmitted to other entities due to interconnectedness and correlation effects and contributed disproportionately to systemic losses during the crisis. The effect was exacerbated due to the absence of macro-prudential supervision and controls in most countries. Monitoring financial stability through the implementation of macro-prudential policies thus became an important objective of supervisors after the crisis (Kahou and Lehar 2017).

Furthermore, in the aftermath of the GFC, governments in Europe and the United States (us) had to use mergers and other bailout mechanisms for many of the distressed banks. More recently, in March 2023, the collapse of Silicon Valley Bank (svb), a specialised bank for tech start-ups in the us, and Credit Suisse, a major Swiss bank, also triggered regional crises. Through regulatory intervention, both banks were later acquired by other banks. While such crisis-led bank consolidations avoid the burden of fiscal rescue measures, they can, in turn, create potentially riskier mega banks and a more concentrated bank market, which can change the dynamics of systemic vulnerability (De Nicoló et al 2004). This concern has highlighted the need for an in-depth evaluation of the consolidation-stability nexus even at a national level.

The Indian banking sector experienced a period of banking crisis during 2012–15, engendered by worsening asset quality and burgeoning losses, especially among public sector banks (psbs). Subsequently, the Government of India initiated major reforms for psbs from 2016 onwards. As part of the agenda, several weak psbs were merged with their relatively stronger counterparts between 2017 and 2020. Even as the last of the mergers were achieved, the COVID-19 pandemic (2020–21) stressed both the global and domestic financial markets. This historical landscape has created a unique natural experiment of crises and mergers in India. In this context, the objective of our study is to assess the impact of consolidation and concentration on the financial stability of India's banking sector.

We use Merton's (1974) contingent claims analysis (CCA) to design a novel bank-level metric—marginal risk contribution (MRC)—which captures the extent to which a bank's presence

The authors would like to thank Sanjay Basu for his valuable comments. They also acknowledge the insightful comments and suggestions of the anonymous referee.

The views presented here are solely of the authors and do not necessarily represent those of the National Institute of Bank Management. The usual disclaimer applies.

Tasneem Chherawala (tasneem@nibmindia.org) and Gargi Sanati (gargi@nibmindia.org) are with the NIBM, Pune.

affects systemic stability. Our empirical analysis uses quarterly panel data from March 2012 to June 2022 for a mix of merged banks and other large, unconsolidated public and private banks in India. We apply system GMM (Arellano and Bover 1995; Blundell and Bond 1998) and bias-corrected fixed effects (Everaert and Pozzi 2007) models to explain the variations in MRC estimates. To the best of our knowledge, ours is the first study to adopt the market-based CCA to measure banks' contribution to systemic risk for a comprehensive evaluation of the implications of mergers, concentration, and other factors on the banking system's resilience in India.

We find that stressed mergers increase financial instability. Additionally, banks' idiosyncratic insolvency risk, higher credit-to-deposit ratio, lower capital adequacy, and poor asset quality also substantially weaken the financial system. Furthermore, ownership characteristics also matter in determining systemic risk contribution as PSBs in general and those with higher government shareholding create greater destabilising effects. On the other hand, bank size, a higher value of which reflects greater concentration in the banking sector, does not have a significant influence on systemic risk. Domestic systemically important banks (DSIBs), which are subject to stricter prudential norms, enhance the resilience of the financial system. Thus, broadly, the Indian evidence supports the consolidation-fragility relationship that arises due to mergers but is neutral on concentration-fragility implications associated with bank size. Our results point towards important macro-prudential policy and regulatory measures to guide future actions on bank mergers.

Literature Review

There is a vast body of academic literature that has, over the years, deliberated on the competing paradigms of competition fragility (alternately, concentration stability) and competition stability (alternately, concentration fragility) in banking systems. The competition-fragility relation was first proposed by Keeley (1990), who demonstrated with US data that in a more competitive environment, banks' market power was reduced, in turn increasing bank fragility through worsening default risk and capital cover. This viewpoint was challenged by Mishkin (1999), who advocated the concentration-fragility nexus by arguing that a less competitive banking industry, with larger institutions, could increase systemic risk due to the moral hazard associated with the implicit protection of too-big-to-fail entities.

Subsequently, various opposing perspectives on the impact of competition and concentration on bank risks emerged through empirical and theoretical research. De Nicoló et al (2004) used the z-score methodology on cross-country data for 1995 and 2000 to show that larger financial conglomerates had higher standalone risks and greater concentration of banks translated to higher country-level systemic risk. On the other hand, Beck et al (2006) also used multi-country data between 1980 and 1997 to counter that financial crisis occurred with a lower likelihood for more concentrated banking systems. The early literature surrounding this debate indicated that the relationship between concentration and stability was complex,

multifaceted, and ambiguous (Berger et al 2004; Allen and Gale 2004).

After the GFC, new research on concentration fragility focused on the mergers of distressed banks and whether they increased systemic risk (consolidation fragility) or reduced it (consolidation stability). Vallascas and Hagendorff (2011) analysed post-GFC European bank consolidations using CCA to show that mergers motivated by regulatory incentives increased the default risk of acquiring banks. Applying the Marginal Expected Shortfall ([MES] Acharya et al 2017) to a data set of international, domestic, and cross-border mergers, Weiß et al (2014) evidenced that financial instability increased following mergers and attributed its cause to moral hazard risk rather than bigger bank size.

Maslak and Senel (2023) provided support for consolidation stability using popular measures of systemic risk (like MES; SRISK by Brownlees and Engle 2017; ΔCoVaR by Adrian and Brunnermeier 2016). For US bank mergers during the GFC, they found that market-adjusted systemic risk decreased for the acquirers irrespective of whether the mergers were government-assisted or privately arranged. However, their results pertaining to the influence of large banks on financial stability were inconsistent for different systemic risk metrics.

Van Dellen et al (2018) and IJtsma et al (2017) demonstrated that bank mergers were risk-neutral, but the former cautioned that there could be substantial scope for an increase in the likelihood of acquirers' effect on stability if potential risk reduction benefits did not materialise. Calice et al (2021) applied several mediators of concentration on cross-country data between 1997 and 2015 to conclude that the increasing concentration because of mergers has a non-monotonic impact on financial stability, that is, financial systems remain resilient when concentration increases below a threshold but become fragile if concentration rises above a threshold.

Studies pertaining to the Indian banking sector, which explore the relationship between consolidation, concentration, and systemic risk, are limited. An event study by Jayadev and Sensarma (2007), covering bank acquisitions in India till 2007, suggested that when mergers took place through regulatory intervention, shareholders of neither the bidder nor the target bank benefited. Employing data of Indian banks for 1992–2012, Ghosh (2013) demonstrated that state-owned banks underperformed on a number of parameters as compared to private banks, after the imposition of stringent macroprudential norms. He found that mergers had a positive impact on bank profitability but did not influence bank-specific z-score and credit growth.

Verma et al (2019) applied the Tail-Event driven Network model to capture systemic risk due to interconnectedness among banks but did not consider the impact of mergers. Based on the amalgamation of the State Bank of India with its subsidiaries, Krishnamurthy (2017) highlighted the need for policymakers to carefully plan the subsequent rounds of crisis-affected bank mergers. Das and Kumbhakar (2022) analysed the post-crisis bank consolidations in India to show that cost efficiencies did not improve subsequently and had the potential

to weaken the banking system. Gupta and Kashiramka (2021) indicated that PSBs contributed more to systemic risk as measured by SRISK than private banks, but did not address the implication of mergers.

From the limited articles described earlier, it is clear that the consolidation-fragility hypothesis has not been rigorously tested in the Indian context. This is an important gap that our study tries to fill. We add to the extant literature in a number of ways. To begin with, ours is possibly the first study that measures systemic risk in India using CCA and extends it to evaluate the impact of mergers and concentration on financial stability. Second, we introduce the novel metric of MRC to capture the unconditional impact of individual banks' risk on the overall risk of the system. Third, we include both market-based and accounting information to explain the variations in banks' contributions to systemic risk. Finally, unlike previous studies for India, our analysis is enriched by data at a higher frequency (quarterly) and over a longer period that encompasses multiple crises and a series of restructuring mergers.

Mergers in the Indian Banking Sector: Stylised Facts

The period from 2017 to 2020 saw several mergers of state-owned banks in India, bringing their numbers down from 27 to 12 over a period of three years. The PSB mergers were sandwiched between two financial crises—the endogenous non-performing assets (NPA) crisis between 2012 and 2015, and the COVID-19 pandemic-driven financial market stress between 2020 and 2021. The mergers were orchestrated by the Indian government as part of its reform agenda for strengthening and repositioning PSBs, after the sector went through a period of rapidly worsening credit quality and eroding capital (PIB 2016). The objective of the bank amalgamations was to “facilitate consolidation among PSBs to create strong and competitive banks capable of achieving economies of scale and realisation of synergy benefits with wider product and service offering for customers” (PIB 2020).

Before their respective amalgamations, six of the target PSBs were under regulatory constraints imposed by the Reserve Bank of India (RBI) under the prompt corrective action (PCA) framework (PIB 2018), due to their stressed balance sheets.

Table 1 summarises the average financial positions of the acquiring and target PSBs in the pre- and post-merger periods. As seen in the table, target banks were smaller (with a low average asset share) and weaker in terms of higher non-performing loans and lower capital adequacy. The acquiring banks were relatively larger and healthier, with better asset quality and higher capital adequacy ratios before the consolidation. The merged entities were bigger and increased the banking sector concentration as depicted by the jump in their share in total banking sector assets post-merger. Furthermore, around

Table 1: Financial Performance of PSBs—Before and after the Merger

	In the Year of Merger		One-Year post Merger
	Target Banks	Acquiring Banks	Merged Banks
Average share of assets (%)	1.12	6.20	8.34
Average GNPA ratio (%)	15.52	9.96	11.16
Average tier I CAR (%)	7.94	11.13	10.82

Source: RBI database. Authors' computation.

₹3.11 trillion capital was infused by the government in the PSBs between 2017 and 2021, to strengthen their balance sheets and create market confidence in the consolidation process, of which around 60.5% was allocated to the PSBs that were amalgamated.

Analytical Framework for Estimation of MRC

In order to capture systemic risk vulnerability created by banks, we design the dependent variable, MRC. MRC is based on the structural CCA framework developed by Merton (1974) and its application to quantify bank-level solvency risk (Gray et al 2008) and systemic risk (Gray and Jobst 2010).

Bank-level Insolvency Risk and Systemic Risk

Merton's CCA is applied to estimate the risk of default of a firm (or bank in our case), for which the capital structure is assumed to comprise equity and risky debt D , maturing at time T . Thus, at any point in time t , the firm's market value of assets (A_t) is the sum of the market value of equity E_t and the market value of debt D_t , that is:

$$A_t = E_t + D_t \quad \dots (1)$$

A_t is stochastic and can potentially decline below D (often referred to as the distress barrier or default point) at time T . Since equity represents a contingent claim on the residual value of the firm's assets at maturity, if $A_T > D$, equity holders derive positive value from firm's assets after debt (D) is paid in entirety. This represents the non-default state. On the other hand, if $A_T < D$, the debt will not be repaid, leading to default by the firm, that is, insolvency of the bank. Equity investors thus hold a call option on the market value of assets A_t as the underlying and D as the strike price. At maturity, they receive the maximum of either asset value less the debt or nothing (as shown in equation [2]).

$$E_T = \text{Max}(A_T - D, 0) \quad \dots (2)$$

If the market value of assets falls, the value of the call option, measured by the market value of equity E_t , decreases and the firm's risk of default increases. Based on this rationale, a bank's distance-to-default at time t (DD_t) can be estimated (Gray and Jobst 2010). DD_t measures the number of standard deviations the bank's asset value is away from the default point. The standalone probability of default (PD_t) can be derived from DD_t using the normal distribution transformation given in equation (3).

$$PD_t = N(-DD_t) \quad \dots (3)$$

where $N(\cdot)$ is the cumulative normal distribution.

For an individual bank, a lower value of the distance-to-default translates to a higher value of PD_t , thereby implying higher standalone insolvency risk.

In order to estimate PD_t , we need information on the market value of equity (E_t), the volatility in the market value of equity (σ_E), and the default point (D). E_t can be measured by market capitalisation; σ_E is derived as the annualised standard deviation of stock returns; and D can be estimated by the book value

of outside liabilities (comprising deposits and non-RBI market borrowings) for the bank. Assuming the maturity of the debt obligation to be one year leads to a one-year forward-looking measure of PD .

An alternative metric of standalone bank risk is Z -score (De Nicoló et al 2004), which is derived from accounting values of return on assets (ROA), volatility of asset returns (σ_{ROA}), and the ratio of equity (E) to assets (A) (equation [4]). The logarithmic value of $1+Z$ -score is equivalent to an accounting measure of distance-to-default and is inversely related to the standalone risk of insolvency of banks.

$$Z - \text{Score} = \frac{ROA + \frac{E}{A}}{\sigma_{ROA}} \quad \dots (4)$$

The CCA can be extended to measure systemic default probability. The system is defined as a composite of all banks, where the system-level market capitalisation (E_t^{Sys}) and default threshold (D_t^{Sys}) are the respective aggregates of the constituent banks. The system-level equity volatility (σ_A^{Sys}) is estimated by the standard deviation of bank index returns, where the bank index is constructed from the weighted average equity prices of the component banks.

The systemic default probability is derived as

$$PD_t^{Sys} = N(-DD_t^{Sys}) \quad \dots (5)$$

PD_t^{Sys} can be interpreted as a time-varying measure of systemic risk. The higher its value, the greater the instability of the financial system. An important feature of this measure is that as long as banks' asset returns are imperfectly correlated, the system-level PD will always be lower than the weighted average PD of the constituent banks. The systemic PD will therefore implicitly incorporate the diversification effect across banks, which, depending upon the level of correlation, can increase or reduce the financial stability of the system.

Bank-level Marginal Risk Contribution

Given our definition of standalone and systemic PD , we can now measure the extent to which constituent banks impact systemic risk as a top-down measure of MRC . The MRC for each bank is backed out by examining the difference between the systemic PD with and without that bank.

Thus, the MRC of the i -th bank to the system at time t is given by

$$MRC_{it} = PD_t^{Sys-i} - PD_t^{Sys} \quad \dots (6)$$

MRC identifies ex post the extent to which individual banks increase or reduce financial instability. If MRC is positive for the i -th bank, it implies that the system without the i -th bank is riskier in terms of a higher systemic PD . Thus, the presence of the i -th bank reduces systemic risk and stabilises the financial system. A negative value of MRC implies that the i -th bank increases systemic risk, and its inclusion makes the financial sector more vulnerable.

In this context, it is important to distinguish MRC from other popular market-based measures of systemic risk like MES , $SRISK$, and $\Delta CovAR$. First, the MRC derivation is based on

the sound theory of contingent claims as opposed to the parametric approach followed by the other measures. Second, MRC is derived from the unconditional probabilities of standalone and systemic distress and does not presuppose the occurrence of an extreme shock on the market returns of the system (as in MES and $SRISK$) or of any individual bank (as in the case of $\Delta CovAR$). As such, MRC can be applied both during crisis periods and tranquil times. Third, in using the probability metric rather than a predicted loss metric, we avoid the assumption of a percentile to determine the extreme loss or minimum capital requirement, which is required for the other measures.¹

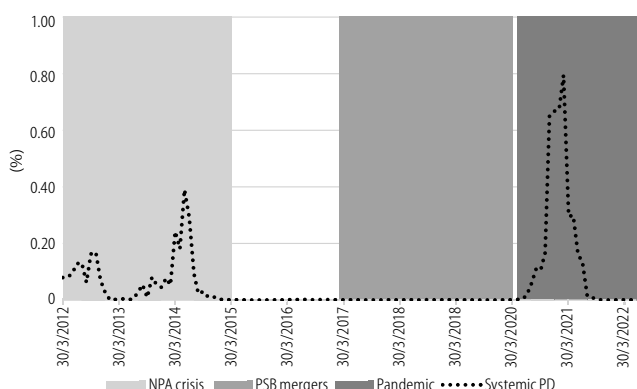
Data and Methodology

We define the Indian financial system using a sample of 24 large domestic commercial banks from March 2012 to June 2022. Before the mergers, this sample represents 77.74% of total banking sector assets as of 31 March 2017. After all the mergers, the sample size is reduced to 13 banks, representing 79.24% of banking sector assets as on 31 March 2022. This sample includes all the banks that were amalgamated between 2017 and 2020 and a control group of large private banks and unconsolidated large $PSBs$.² Small private sector banks, non-merging small $PSBs$, and foreign banks have been excluded due to their minimal implications for systemic risk.

To measure MRC , quarterly closing stock price and market capitalisation data of each bank are taken from March 2012 to June 2022 from the National Stock Exchange (NSE) database. Annualised volatility of equity returns is estimated for the period March 2012 to June 2022, using the preceding 12 months' stock returns. Quarterly data for all other balance sheet and regulatory ratios are taken from the RBI database.³

Next, we discuss the explanatory variables used in our analysis. The Herfindahl-Hirschman index or the market share of top " n " banks is often used as a measure of bank concentration (De Nicoló et al 2004). However, this measure is more meaningful in the context of cross-country analyses over longer periods of time. Since our study focuses on the national banking sector, we use the more direct measure of the log of total assets (TA) to represent size-based concentration, which can increase both due to asset growth and mergers. Since TA makes the impact of asset growth and mergers fungible, we explicitly include the binary variable $MERGED$ to separate the risk implication of bank amalgamation. In doing so, we make a nuanced distinction between the concentration, which can occur when TA increases, and consolidation, which happens with mergers.

$Zscore$ measures the standalone insolvency risk of banks. CAR_{t1} represents the bank's soundness in terms of the availability of good quality (regulatory Tier 1) capital to absorb losses. $GNPA$ measures the bank's ex post credit risk. The CD ratio captures the bank's liquidity risk in terms of the availability of stable sources of funds to finance loans. $DSIB$ is the regulatory indicator of domestic systemic importance based on the RBI's classification. The extent of government ownership for each bank has been captured by the variable $GOIS$. $GDPGR$ captures the macroeconomic cycle.

Figure 1: Trend in Systemic PD of the Indian Banking Sector

Source: NSE Equity Data and RBI database, Authors' computation.

Table 2 defines the variables, we have used for our empirical analysis.

The descriptive statistics of our bank-level variables are provided in Table 3. The mean *MRC* is 0.003%, implying that on average, banks in the sample have reduced systemic vulnerability. The mean Tier 1 capital adequacy (*CAR_{T1}*) at 11.212% has been higher than the regulatory minimum. The averages for the *GNPA* ratio and *CD* ratio are 6.783% and 77.587%, respectively, and both have higher variability. The

Table 2: Variable Definitions and Data Sources

Variable	Definition	Source
Dependent variable		
<i>MRC</i>	Individual bank's MRC to systemic risk	Authors' calculation, NSE database
Explanatory variables		
<i>MERGED</i>	Bank-wise binary variable, which takes the value 1 if the bank has been amalgamated and 0 otherwise	Public information
<i>Z-score</i>	Measure of standalone insolvency risk, defined earlier	Authors' calculation
<i>TA</i>	Natural logarithm of the total assets of the bank	RBI database
<i>MRCL1</i>	One-quarter lagged value of MRC	Authors' calculation
<i>CAR_{T1}</i>	Tier 1 Capital Adequacy Ratio = $\frac{\text{Tier 1 Regulatory Capital}}{\text{Risk Weighted Assets}}$	RBI database
<i>CD</i>	Credit to deposit ratio	
<i>GOIS</i>	Share of government holding in the bank's equity	
<i>GNPA</i>	Gross non-performing loans as a ratio of total loans	
<i>DSIB</i>	Dummy variable which takes the value 1 for domestic systemically important banks as classified by the RBI and 0 otherwise	
<i>GDPGR</i>	Quarterly growth rate of India's gross domestic product	

Source: Authors' compilation.

Table 3: Summary Statistics of Bank-level Variables (All Banks)

	<i>MRC</i>	<i>CAR_{T1}</i>	<i>CD</i>	<i>GNPA</i>	<i>TA</i>	<i>GOIS</i>	<i>Zscore</i>
Mean	0.003	11.212	77.587	6.783	8.602	43.617	2.953
Std dev	0.060	3.304	13.572	5.115	0.803	35.604	0.894
Skewness	2.235	0.321	0.912	0.790	0.112	-0.284	-0.076
Kurtosis	55.824	4.702	6.380	2.663	3.480	1.324	3.762
No of obs	546	546	546	546	546	546	546

All mean and standard deviation values are in percentage except *TA* and *Z-score*.

Source: Authors' estimation.

high value of the mean *Zscore* suggests that broadly, the standalone risk of banks has been relatively low.

Our bank-level empirical model is constructed to estimate the impact of bank mergers and other variables on bank-wise systemic risk contribution, as measured by *MRC*, as the dependent variable. As per the definition of *MRC*, a positive coefficient for an explanatory variable in our model implies that higher values of the variable make *MRC* less negative or more positive and thereby reduce the vulnerability of the system due to the presence of the bank.

We explain the variations in *MRC* estimates using system GMM (Arellano and Bover 1995; Blundell and Bond 1998) and bias-corrected fixed effects/random effects (Everaert and Pozzi 2007) models. It may be noted that endogeneity bias may be captured by using GMM estimators (refer to Arellano and Bover 1995; Blundell and Bond 1998). We run two-step system GMM models that fulfil all three additional conditions: the presence of significant AR (1) correlation, the absence of significant AR (2) correlation, and valid over-identifying conditions.

Everaert and Pozzi (2007) use the bootstrap correction, which maintains high efficiency compared to GMM and rectifies the bias in the least squares dummy variable (LSDV) estimates. This is known as a bias-corrected fixed-effects/random-effects model (BCFE/BCRE) model in the literature. The BC estimator has been further improved with finite-sample properties (Breitung et al 2022). Since the analytical form of the bias is known, the BC estimator can correct it directly at the source by adjusting the respective moment conditions. Moreover, the BC model addresses the problem of higher-order autoregressive models, and as the BC estimator is a method of moments estimator with known asymptotic distribution, the standard errors can be readily computed. Following the specification of Everaert and Pozzi (2007) and Zaman et al (2022) we represent the bias-corrected (BC) estimators as follows

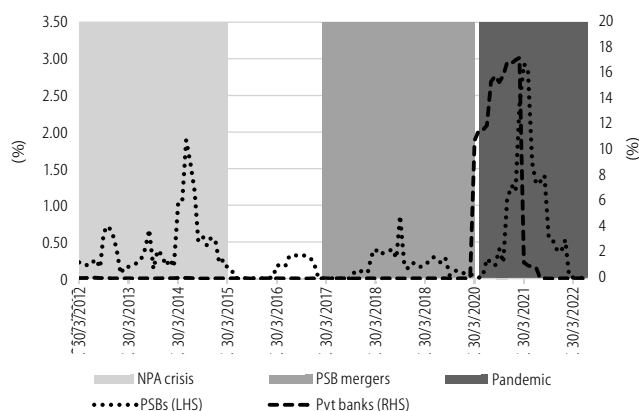
$$MRC_{it} = \alpha_0 + \alpha_1 MRC_{it-1} + \beta' X_{it} + a_i + \epsilon_{it} \quad \dots (6)$$

where ϵ_{it} denotes a random term, a_i is the unobserved heterogeneity and MRC_{it-1} is the *MRC* estimate of bank *i* at period *t-1*. *X* denotes the set of explanatory variables.

Empirical Results

In this section, we discuss our empirical results, in which we depict the trend of systemic PD and bank group-wise differences in *MRC* and discourse upon the results of the econometric analysis.

Trends in systemic risk and MRC: Systemic PD, derived as per the methodology described earlier, is plotted over time (Figure 1) and shows elevated levels during the periods of the NPA crisis (March 2012 to March 2015) and COVID-19 crisis (March 2020 to March 2021). The financial shock during the pandemic was more intense and of relatively shorter duration as compared to the NPA crisis. There was a period of relative calm in the financial markets between the two crises, as depicted by the close-to-zero value of the systemic PD between March 2015 and March 2020.

Figure 2: Systemic PD of Public Sector Banks versus Private Banks

Source: NSE Equity Data and RBI database. Authors' computation.

Table 4: Average MRC of Banks in the Sample (%)

	NPA Crisis (2012–15)	Pandemic Crisis (2020–21)
Target banks	-0.001	
Acquiring banks	-0.004	-0.012
Unconsolidated private banks	0.030	0.018
Unconsolidated PSBs	-0.005	-0.005

Source: NSE Equity Prices and RBI database. Authors' computation.

PSBs and private banks were adversely impacted during both crises. However, the nature and extent of systemic stress within the two banking groups were distinctly different. As shown in Figure 2, the systemic PD of PSBs had greater variability and was higher than that of private banks during the NPA crisis. However, the private bank group was more stressed during the pandemic. Diversification benefits across private and public sector bank groups during both crises were evident as the values of overall systemic PDs (Figure 1) were much lower than those of either of the standalone banking groups.

Table 4 summarises the average MRCs by various banking groups. As can be seen, all acquiring and target banks hurt financial stability, with negative MRCs on average under the NPA crisis. Also, the contribution of the acquirer banks to systemic risk was higher post their merger during the pandemic crisis as compared to the pre-merger financial shock. On the other hand, the private banks had collectively reduced systemic risk through positive average MRCs in both crises. This indicates support for the consolidation-fragility hypothesis for the Indian banking sector.

Econometric estimation: We summarise the findings of our econometric analyses of BCRE estimation in Table 5 and two-stage system GMM (Table 6) in this section. It may be noted that an increase in the value of MRC reduces systemic vulnerability contributed by the bank.

The results of our empirical model (Table 5, M1 and M2; Table 6 M3) show that mergers

have a robust and strong negative impact on MRC as depicted by the coefficient for the variable *MERGED*, which is significant at the 1% level (M1 and M3) and 5% level (M2). So, the mergers have increased financial fragility, and thus, the consolidation-fragility relationship is borne out in the Indian context.

Additionally, the contribution to systemic risk is adversely impacted by the standalone insolvency risk of banks as depicted by the positive and significant (at the 1% level) coefficient of *Zscore* in M1 (Table 5) and M5 (Table 6). Thus, banks that have a higher z-score and therefore lower individual risk of default reduce systemic vulnerability.

Table 5: Bias-corrected Random Effect Model

	M1	M2
<i>MRCL1</i>	0.721973*** (0.009453)	0.711037*** (0.007374)
<i>MERGED</i>	-0.003244*** (0.001043)	-0.003239** (0.001492)
<i>CAR_t1</i>	0.000299*** (0.000113)	0.000406** (0.000167)
<i>CD</i>	-0.000392*** (0.000058)	-0.000359*** (0.000118)
<i>GNPA</i>	0.000001 (0.000187)	-0.000402** (0.0001872)
<i>GOIS</i>	-0.000089*** (0.00003)	-0.000066* (0.000035)
<i>TA</i>		0.000451 (0.000499)
<i>GDPGR</i>	-0.000774 (0.000694)	
<i>Zscore</i>	0.003990*** (0.000811)	
<i>_CONS</i>	0.021414** (0.008513)	0.025408* (0.013723)
No of Obs	533	533

***, **, and * represent significance level at 1%, 5%, and 10%, respectively. Standard errors reported in parenthesis. We have reported the best-fitted model. All numbers are reported up to six decimal points.

Source: Authors' estimation.

Table 6: Two-stage Arellano-Bover/Blundell-Bond Estimation

	M3	M4	M5	M6
<i>MRCL1</i>	0.730715 *** (-0.011791)	0.680599 *** (-0.018606)	0.690660 *** (-0.024708)	0.696654 *** (-0.005889)
<i>MERGED</i>	-0.003495 ** (0.00175)	0.001216 (0.002497)	0.000742 (0.003036)	-0.000061 (0.00375)
<i>CAR_t1</i>	0.000654 *** (0.0001209)	0.000569 *** (0.000203)	0.00033 *** (0.000082)	-0.0003 (0.000214)
<i>CD</i>	-0.000723 *** (0.00038)	-0.000648 *** (0.000055)	-0.000541 *** (0.00004)	-0.000772 *** (0.000059)
<i>GNPA</i>	-0.001434 *** (0.000271)	-0.000462 (0.000547)		-0.000982 ** (0.000411)
<i>TA</i>	-0.002895 (0.003618)	-0.00199 (0.003063)	-0.002995 (0.003857)	
<i>GDPGR</i>		-0.000323 (0.000479)		
<i>Zscore</i>			0.007233 *** (0.000902)	
<i>GOIS</i>				0.000511 (0.000462)
<i>DSIB</i>				0.120673 *** (0.041048)
<i>_CONS</i>	0.083944 *** (0.030829)	0.064374 ** (0.028094)	0.042298 (0.035427)	0.014283 (0.034025)
No of obs	533	533	533	533
Diagnosis test				
AR(2)	-1.7022 (p=0.0887)	-1.6325 (p=0.1026)	1.6418 (p=0.1006)	1.6818 (p=0.0926)
Sargan	chi2(480)=10.71	chi2(480)=5.16	chi2(480)=5.89	chi2(480)=6.38

***, **, and * represent significance level at 1%, 5%, and 10%, respectively. Standard errors are reported in brackets. We do not reject the null hypothesis of the Sargan test as overidentifying restrictions are valid, as the probability of error is more than 99% for all the reported Sargan test results.

Source: Authors' estimation.

We do not reject the null hypothesis of the Sargan test as over-identifying restrictions are valid, as the probability of error is more than 99% for all the reported Sargan test results.

Drilling down to the risk profile of individual banks in terms of credit risk (measured by *GNPA*) and liquidity risk (measured by *CD*), we see that both have a significant adverse impact on systemic risk. The *GNPA* ratio has a negative coefficient, which is significant at the 1% (*M3*) and 5% level (*M2* and *M6*). Larger loan losses can weaken a bank and increase its threat to the system. The coefficient of the *CD* ratio, in particular, is negative and significant at the 1% level across all models. An increasing *CD* ratio typically occurs when credit growth outpaces deposit growth, implying lower stable funding and greater liquidity stress for the bank, which can contribute to systemic stress.

Furthermore, the variable *Car_t1* consistently has a strong positive impact on *MRC*. Thus, banks that have maintained adequately high levels of good quality regulatory capital have a greater stabilising effect on the system since they are better positioned to absorb unexpected future losses and withstand adverse financial shocks. This justifies the government's injection of capital for the merging banks to strengthen their balance sheets and alleviate potential systemic risk during and after the merger.

Similar to the conclusion of Gupta and Kashiramka (2021), the negative and statistically significant coefficient for *GOIS* (at the 1% level in *M1* and the 5% level in *M2*) suggest that *PSBs*, and especially those with higher shareholding by the government, created greater vulnerability in the financial system as opposed to private banks. Government shareholding in *PSBs* typically increases due to recapitalisation of those banks which may otherwise fail since they cannot raise capital from the market, and such bail-outs do not bode well for financial stability. This result corroborates the moral hazard risk of government ownership in the context of systemic vulnerability impact.

Interestingly, the variable *TA* is not statistically significant in any of the models. Thus, bigger banks do not necessarily increase financial vulnerability. Our results therefore indicate a neutral stance on the concentration-fragility paradigm. The positive significant coefficient of *DSIB* (Model 6) shows that the larger banks, classified as systemically important by the RBI, have enhanced overall systemic resilience. It may be argued here that this may be because of the higher capital buffers imposed upon the *DSIBs*, which lead to a beneficial impact on financial stability.

Finally, as can be seen in *M1* (Table 5) and *M4* (Table 6), the introduction of the variable *GDPGR*, which represents the macroeconomic cycle, does not have any significant influence on banks' marginal risk contributions.

The combined results suggest that stressed mergers are destabilising for the financial system. However, the impact on systemic risk is primarily influenced by banks' standalone soundness parameters like liquidity, non-performing loans ratio, and capital adequacy rather than size. On the other hand, financial stability is neutral to bank size and therefore banking sector concentration and systemically important banks

may enhance financial stability, subject to lower standalone risk and stricter regulatory norms. This suggests that the adverse effect of stressed mergers on systemic risk can be offset if the larger merged entity is subject to more stringent micro-prudential criteria. Thus, our results support the consolidation-fragility nexus but not the concentration-fragility relationship.

Conclusions

In this paper, we explore the nuanced implications of bank consolidation due to mergers and size-based concentration on systemic risk in light of two periods of financial crises and a wave of government-led bank mergers that have occurred over the last decade in India. We also examine the role of other bank-specific factors as drivers of financial stability.

Using a bank-level, market-based measure of marginal risk contribution, we find that the consolidation of stressed banks increases the systemic risk contributions of acquiring entities. This result supports the consolidation-fragility nexus in India. The destabilising effect is also strongly influenced by the standalone insolvency risk, bank-wise credit and liquidity risks, and moral hazard risk of government ownership. Systemic vulnerability can, however, be alleviated by higher capital adequacy of banks. On the other hand, concentration based on bank size does not significantly contribute to systemic risk. Moreover, banks that are deemed to be systemically important and subject to stricter prudential norms can mitigate systemic risk during a crisis.

Our results point towards a number of important policy implications. First, enforcing weak bank mergers through regulatory intervention may not be the best policy option from the perspective of financial stability. However, if such a resolution mechanism is considered going forward, regulatory oversight should focus on minimising systemic vulnerabilities through better ex ante risk profiles of acquiring banks. Particularly, for stressed mergers, government capital infusion may be unavoidable to improve the capital adequacy of the weak banks and boost investor confidence. However, this support should be for a limited period till the merged entity stabilises.

Second, taking a cue from the Dodd-Frank Act enacted in 2010 in the US, macro-prudential policies for the financial sector may include mandatory assessment of the systemic risk impact of proposed mergers between banks and even non-bank acquisitions. Alternatively, higher capital requirements can be imposed on the merged entity, which internalises the bailout cost and offsets adverse financial shocks (Ma and Nguyen 2021).

Third, defining systemically important banks primarily by size may not suffice, and micro-prudential norms must continue to emphasise appropriate risk controls for all financial institutions. In highly interlinked financial markets, the collapse of even smaller, less regulated banks can trigger contagion-driven financial instability, as was seen when the *svb* failed in the US.

Finally, systemic vulnerability is higher for banks with larger government shareholding. Thus, *PSBs* should be encouraged to raise incremental capital from the markets. This can strengthen market discipline and improve the overall efficiency and financial stability of the banking sector.

NOTES

- 1 Pampurini and Quaranta (2025) provide a detailed comparison of CCA and other market-based systemic risk measures.
- 2 Treated banks (acquirer): State Bank of India, Bank of Baroda, Canara Bank, Punjab National Bank, Union Bank of India and Indian Bank. Treated banks (target): SBI subsidiaries (SBBJ, SBT and SBM), Dena Bank, Vijaya Bank, Syndicate Bank, Oriental Bank of Commerce, United Bank of India, Andhra Bank, Corporation Bank and Allahabad Bank. Control banks (non-merged): Axis Bank, HDFC Bank, ICICI Bank, Kotak Bank, Yes Bank, Bank of India and Central Bank of India.
- 3 Database on Indian Economy; Banking-Performance Indicators; Variables to be published Bank and Bank Group wise (For Public Access), <https://data.rbi.org.in/#/dbie/reports/Statistics/Financial%20Sector/Banking%20-%20Performance%20Indicators>.

REFERENCES

- Acharya, V V, L H Pedersen, T Philippon and M Richardson (2017): "Measuring Systemic Risk," *Review of Financial Studies*, Vol 30, No 1, pp 2–47.
- Adrian, T and M K Brunnermeier (2016): "CoVaR," *American Economic Review*, Vol 106, No 7, pp 1705–41.
- Allen, F and D Gale (2004): "Competition and Financial Stability," *Journal of Money, Credit and Banking*, Vol 36, No 3, Part 2: Bank Concentration and Competition: An Evolution in the Making a Conference Sponsored by the Federal Reserve Bank of Cleveland, 21–23 May 2003, pp 453–80.
- Arellano, M and O Bover (1995): "Another Look at the Instrumental Variable Estimation of Error-Component Models," *Journal of Econometrics*, Vol 68, pp 29–52, [http://dx.doi.org/10.1016/0304-4076\(94\)01642-D](http://dx.doi.org/10.1016/0304-4076(94)01642-D), viewed on 28 May 2024.
- Beck, T, A Demirci-Kunt and R Levine (2006): "Bank Concentration, Competition, and Crises: First Results," *Journal of Banking and Finance*, Vol 30, pp 1581–1603.
- Berger, A N, A Demirci-Kunt, R Levine and J G Haubrich (2004): "Bank Concentration and Competition: An Evolution in the Making," *Journal of Money, Credit and Banking*, Vol 36, No 3, Part 2: Bank Concentration and Competition: An Evolution in the Making a Conference Sponsored by the Federal Reserve Bank of Cleveland, 21–23 May 2003, pp 433–51.
- Blundell, R and S Bond (1998): "Initial Conditions and Moment Restrictions in Dynamic Panel Data Models," *Journal of Econometrics*, Elsevier, Vol 87, No 1, pp 115–43.
- Breitung, J, S Kripfganz and K Hayakawa (2022): "Bias-corrected Method of Moments Estimators for Dynamic Panel Data Models," *Econometrics and Statistics*, Elsevier, Vol 24(C), pp 116–32.
- Brownlees, C and R Engle (2017): "SRISK: A Conditional Capital Shortfall Measure of Systemic Risk," *The Review of Financial Studies*, Vol 30, No 1, pp 48–79.
- Calice, P, L Leonida and E Muzzupappa (2021): "Concentration-Stability vs Concentration-Fragility: New Cross-country Evidence," *Journal of International Financial Markets, Institutions and Money*, Elsevier, Vol 74(C).
- Das, A and S C Kumbhakar (2022): "Bank Merger, Credit Growth and the Great Slowdown in India," *Economic & Political Weekly*, Vol LVII, No 20.
- De Nicolò, G, P Bartholomew, J Zaman, M Zephirin (2004): "Bank Consolidation, Internationalization, and Conglomeration: Trends and Implications for Financial Risk," *Financial Markets, Institutions and Instruments*, Vol 13, pp 173–217.
- Everaert, G and L Pozzi (2007): "Bootstrap-based Bias Correction for Dynamic Panels," *Journal of Economic Dynamics and Control*, Vol 31, pp 1160–84.
- Ghosh, S (2013): "Macroprudential Regulation and Bank Performance: Does Ownership Matter?" *Journal of Banking Regulation*, Vol 16, pp 22–38.
- Gray, D, A Jobst (2010): "Systemic Contingent Claims Analysis—Estimating Market-Implied Systemic Risk," IMF Working Paper, Monetary and Capital Markets Department, WP/13/54, <https://www.imf.org/external/pubs/ft/wp/2013/wp1354.pdf>, on 29 May 2024.
- Gray, D, R Merton and Z Bodie (2008): "A Contingent Claims Analysis of the Subprime Credit Crisis of 2007–2008," World Scientific Reference on Contingent Claims Analysis in Corporate Finance, Volume 4: Contingent Claims Approach for Banks and Sovereign Debt, <https://doi.org/10.1142/9857-vol4>, viewed on 29 May 2024.
- Gupta, J and S Kashiramka (2021): "Identification of Systemically Important Banks in India Using SRISK," *Journal of Financial Regulation and Compliance*, Vol 29, No 4, pp 387–408.
- Ijtsma, P, L Spierdijk and S Shaffer (2017): "The Concentration-stability Controversy in Banking: New Evidence from the EU-25," *Journal of Financial Stability*, Vol 33, pp 273–84.
- Jayadev, M and R Sensarma (2007): "Mergers in Indian Banking: An Analysis," *South Asian Journal of Management*, Vol 14, No 4, pp 20–49.
- Kahou, M E and L Lehar (2017): "Macroprudential Policy: A Review," *Journal of Financial Stability*, Vol 29, pp 92–105.
- Keeley, M C (1990): "Deposit Insurance, Risk and Market Power in Banking," *American Economic Review*, December, Vol 80, No 5, pp 1183–1200.
- Krishnamurthy, R (2017): "Public Sector Bank Mergers: A Reality Check," *Economic & Political Weekly*, Vol LII, No 29.
- Ma, C and X-H Nguyen (2021): "Too Big to Fail and Optimal Regulation," *International Review of Economics and Finance*, Vol 75, pp 747–58.
- Maslak, G D and G Senel (2023): "Bank Consolidation and Systemic Risk: M&A during the 2008 Financial Crisis," *Journal of Financial Services Research*, Vol 63, No 2, pp 201–20.
- Merton, R (1974): "On the Pricing of Corporate Debt: The Risk Structure of Interest Rates," *Journal of Finance*, Vol 29, pp 449–70.
- Mishkin, F S (1999): "Financial Consolidation: Dangers and Opportunities," *Journal of Banking and Finance*, Vol 23, pp 675–91.
- Pampurini, F and A G Quaranta (2025): "Measuring Systemic Risk: A Review of the Main Approaches," *Systemic Risk and Complex Networks in Modern Financial Systems*, V Pacelli (ed), New Economic Windows, Springer, Cham, https://doi.org/10.1007/978-3-031-64916-5_11.
- PIB (2016): "Mergers of Nationalised Banks," Press Information Bureau, Ministry of Finance, Government of India, 19 July 2016, <https://archive.pib.gov.in/>.
- (2018): "Prompt Corrective Action (PCA) of RBI on Public Sector Banks (PSBs)," Press Information Bureau, Ministry of Finance, Government of India, 16 March, <https://archive.pib.gov.in/>.
- (2020): Cabinet Approves Mega Consolidation in Public Sector Banks (PSBs) with effect from 1.4.2020, Press Information Bureau, Ministry of Finance, Government of India, 4 March, <https://archive.pib.gov.in/>.
- Roodman, D (2009): "A Note on the Theme of Too Many Instruments," *Oxford Bulletin of Economics and Statistics*, Vol 71, pp 135–58.
- Vallascas, F and J Hagendorf (2011): "The Impact of European Bank Mergers on Bidder Default Risk," *Journal of Banking and Finance*, Vol 35, No 4, pp 902–15.
- Van Dellen, S, A Benamraoui, H Ngo and J Salaber (2018): "The Effects of Mergers and Acquisitions on Acquiring Banks' Contribution to Systemic Risk," paper presented at the 27th Edition of International Rome Conference on Money, Banking and Finance, Rome, Italy, 10–11 December, <http://westminsterresearch.wmin.ac.uk/>, viewed on 28 May 2024.
- Verma R, W Ahmad, G S Uddin and S Bekiros (2019): "Analysing The Systemic Risk Of Indian Banks," *Economics Letters*, Vol 176, pp 103–08.
- Weiß, GNF, S Neumann and D Bostandzic (2014): "Systemic Risk and Bank Consolidation: International Evidence," *Journal of Banking and Finance*, Vol 40, pp 165–81.
- Zaman, M S, V Valiyattoor and A K Bhandari (2022): "Dynamics of Total Factor Productivity Growth: An Empirical Analysis of Indian Commercial Banks," *Journal of Economic Asymmetries*, Vol 26, pp 2–14.



दी माइक्रोफाइनेंस रिव्यू, खंड-17, संख्या-1, जनवरी-जून 2025

THE MICROFINANCE REVIEW, Volume – XVII, No.1, January-June 2025

शोध पत्र आमंत्रण /CALL FOR PAPERS

बैंकर ग्रामीण विकास संस्थान (बर्ड), लखनऊ नाबार्ड द्वारा प्रवर्तित एक प्रमुख प्रशिक्षण संस्थान है। बर्ड के भीतर स्थापित वित्तीय समावेशन और सूक्ष्मवित्त अनुसंधान केंद्र (सीआरएफआईएम) "दी माइक्रोफाइनेंस रिव्यू" नामक अर्धवार्षिक पत्रिका प्रकाशित कर रहा है। जनवरी-जून 2025, खंड- XVII, संख्या 1 अंक में प्रकाशन के लिए भारत और विदेशों में वित्तीय समावेशन और माइक्रोफाइनेंस क्षेत्र से संबंधित मुद्दों पर शोधपत्र/टिप्पणियां आमंत्रित की जाती हैं।

Bankers Institute of Rural Development (BIRD), Lucknow is a premier training establishment promoted by NABARD. Centre for Research on Financial Inclusion and Microfinance (CRFIM), set up within BIRD has been publishing a half-yearly journal titled "THE MICROFINANCE REVIEW". Papers / commentaries are invited on issues pertaining to the financial inclusion and microfinance sector in India and abroad for publication in the January-June 2025, Volume – XVII, No.1 issue.

शोधपत्र/टिप्पणियां प्रस्तुत करने के लिए विस्तृत दिशानिर्देश इस लिंक पर उपलब्ध है :

Detailed guidelines for submission of papers / commentaries are available at :

<https://birdlucknow.nabard.org/StaticPages/crfim.aspx#MFR-Journals>

इच्छुक लेखक उपरोक्त विषयों पर अपने शोध लेख/टिप्पणियां 30 जून 2025 तक निम्न पते पर प्रेषित कर सकते हैं।

Interested persons may submit papers / commentaries on any of the above themes latest by 30 June 2025.

The Editor, THE MICROFINANCE REVIEW, Bankers Institute of Rural Development, Sector-H, LDA Colony,

Kanpur Road, Lucknow - 226 012; Email: birdjournal@nabard.org / bird@nabard.org;

Tel : 0522-2421119 / 2421136, Website : <https://birdlucknow.nabard.org/>

Director/निदेशक